# ARAVINDHAN S

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## PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing and integrating Machine Learning, Deep Learning, and Generative AI solutions using Python. Experienced in building AI-powered applications, backend APIs, RAG pipelines, agentic workflows, data preprocessing pipelines, and asynchronous AI systems. Strong foundation in machine learning algorithms, model development from scratch, PyTorch, Scikit-Learn, FastAPI, SQL, and AI/LLM integration. Experienced in deploying, testing, monitoring, and troubleshooting AI applications and interested in solving real-world problems through practical AI solutions.

## TECHNICAL SKILLS

**Programming & AI Development**
Python | Object-Oriented Programming (OOP) | AsyncIO | FastAPI | REST APIs | End-to-End System Design

**Machine Learning & Deep Learning**
PyTorch | Scikit-Learn | Neural Networks | Machine Learning Algorithms | Deep Learning | Model Development from Scratch | Model Evaluation | Feature Engineering | Data Preprocessing | Computer Vision | YOLO | Vision Transformers (ViT) | Natural Language Processing (NLP) | Recommender Systems | Graph Neural Networks | GraphSAGE | KNN | Locality Sensitive Hashing (LSH)

**Generative AI**
Large Language Models (LLMs) | Transformers | Retrieval-Augmented Generation (RAG) | RAG Evaluation | Prompt Engineering | Agentic AI Systems | LangChain | LangGraph | CrewAI | Model Context Protocol (MCP) | Fine-Tuning | Vector Databases | FAISS | Chroma

**Data & Analytics**
Data Cleaning | Data Preprocessing | Exploratory Data Analysis (EDA) | Missing Value Handling | Outlier Detection | Feature Scaling | Encoding | Cross-Validation | Statistical Analysis | SQL

**AI Infrastructure & Deployment**
Docker | Celery | Redis | Redis Pub/Sub | Distributed Task Queues | Event-Driven Architecture | Asynchronous Inference | Git | GitHub Actions | Pytest | Prometheus

**Databases**
PostgreSQL | MySQL | SQLAlchemy | Alembic | Vector Databases

## WORK EXPERIENCE

### AI Systems Engineer

**Vankaal**

*02/2026 - 08/2026*

- Developed a real-time AI voice consultation system using Python, FastAPI, and Twilio for automated technical support.
- Integrated LLM-based communication capabilities with custom deterministic planning and business-logic components to maintain controlled AI decision-making.
- Built asynchronous backend workflows for handling real-time voice processing, conversation state, context tracking, and AI responses.
- Developed high-throughput WhatsApp automation workflows using the Meta Graph API and custom webhooks.
- Implemented backend state management and multi-tenant communication workflows for reliable processing of concurrent user requests.
- Integrated AI services into application workflows through APIs and asynchronous processing architectures.
- Intelligent Email Automation & Data Extraction: Architected an event-driven AI workflow that categorizes incoming emails, extracts structured information from PDFs and invoices, and generates context-aware responses using NLP pipelines.
- Built preprocessing workflows to transform unstructured documents and incoming data into structured information for downstream AI processing.
- Developed asynchronous processing workflows for handling email, document, and AI-processing tasks.
- Algorithmic Recommender Systems: Designed and implemented personalized recommendation systems using statistical machine learning, KNN, LSH, and graph-based similarity techniques.
- Developed recommendation algorithms from scratch using mathematical and machine learning fundamentals rather than relying entirely on high-level APIs.
- Evaluated similarity-based recommendation approaches and optimized user-item matching for efficient retrieval.

### Generative AI Developer

**AI Futurix Innovlabz**

*02/2025 - 01/2026*

- Developed Python-based microservices for integrating and serving AI models through backend APIs.

- Built FastAPI services for AI-powered applications using asynchronous processing and REST-based API integration.
- Developed and integrated Generative AI solutions involving LLMs, RAG pipelines, agentic workflows, and AI infrastructure.
- Implemented data and message-processing workflows using AsyncIO, Redis, and asynchronous task architectures.
- Used Docker to package and maintain consistent AI application environments across development and deployment.
- Developed secure API endpoints using OAuth2 authentication and JWT-based authorization.
- Implemented automated testing using Pytest to validate AI application components and backend services.
- Monitored AI application and service performance using Prometheus and investigated application-level issues.
- Worked with AI models and backend systems to integrate machine learning capabilities into practical software applications.

## KEY PROJECTS

### Distributed GenAI Infrastructure & Workflow Engine

*Tech Stack: Python, FastAPI, Celery, Redis, Docker,Postman*

- Architected an asynchronous GenAI pipeline using FastAPI, Celery, and Redis to separate API processing from computationally intensive AI inference.
- Designed distributed AI workers for asynchronous processing and high-concurrency request handling.
- Implemented Redis Pub/Sub for real-time token streaming between AI workers and application services.
- Optimized model and resource management using a singleton worker pattern, allowing AI models, vector databases, and MCP clients to initialize once and remain available across requests.
- Designed fault-isolation mechanisms to prevent AI worker failures from directly affecting API availability.
- Focused on low-latency inference, scalable AI services, asynchronous processing, and production-oriented GenAI infrastructure.

### Autonomous Voice Agent - IVR System

*Tech Stack: Python, CrewAI, Twilio, Whisper, Silero VAD, Redis*

- Architected a low-latency AI voice pipeline combining speech recognition, voice activity detection, LLM reasoning, and text-to-speech components.
- Integrated Silero VAD and Whisper for real-time speech detection and transcription.
- Developed multi-agent conversation workflows using CrewAI for intent handling, reasoning, and conversation state management.
- Integrated Twilio APIs for real-time voice communication and automated customer interactions.
- Implemented asynchronous processing to reduce latency across the speech-to-AI response pipeline.
- Integrated Prometheus monitoring to track real-time system and call metrics.
- Deployed AI voice services on cloud infrastructure.

### Graph Neural Network Recommendation Engine - Built from Scratch

*Tech Stack: Python, NumPy, PyTorch, Linear Algebra, Calculus*

- Implemented a custom GraphSAGE-based recommendation system without relying on high-level graph learning libraries.
- Developed mean aggregation and neighborhood representation functions using linear algebra fundamentals.
- Implemented the backpropagation pipeline to understand gradient flow and optimization at a fundamental level.
- Applied Locality Sensitive Hashing (LSH) for efficient similarity search in high-dimensional spaces.
- Used probability and mathematical concepts for similarity detection and recommendation.
- Focused on implementing and understanding core machine learning algorithms rather than relying entirely on abstraction-heavy frameworks.

## EDUCATION

### Bachelor of Technology / Bachelor's Degree in Artificial Intelligence and Data Science

**Dr. N.G.P. Institute of Technology**

*10/2021 - 04/2025*

GPA: 8.0 / 10

## CORE STRENGTHS

- Machine Learning & Deep Learning
- AI Application Development
- Data Preprocessing & Analysis
- RAG System Development
- Distributed AI Workflows
- Algorithmic Problem Solving
- Generative AI & LLM Integration
- Model Development from Scratch
- AI / ML API Integration
- AI Backend Engineering
- Model Evaluation & Experimentation
- Mathematical Foundations of AI